

Chapter 4

Nonresponse Error

4.1 Introduction

Nonresponse is an error of nonobservation like coverage error (see chapter 5). However, nonresponse error differs from coverage error in that nonresponse reflects an unsuccessful attempt to obtain the desired information from an eligible unit, whereas coverage error (as noted in the next chapter) reflects the failure to have the sample unit uniquely included in the frame.

A great deal of research and attention has been devoted to nonresponse errors (for reviews see Groves 1989; Groves and Couper 1998; Lessler and Kalsbeek 1992). During the last decade, issues of nonresponse error and nonresponse bias in survey estimates have become an important federal interagency survey research topic. In 1997, two interagency work groups, one addressing household surveys and the other addressing establishment surveys, began discussions that focussed specifically on the problem of nonresponse error (Atrostic and Burt 1999; Kydonieffs and Stanley 1999). A previous subcommittee of the FCSM was also devoted to nonresponse in federal surveys; however, that group chiefly focused on trends in nonresponse rates over time (Gonzalez, Kasprzyk, and Scheuren 1995; Shettle et al. 1994; Johnson et al. 1994; and Osmint et al. 1994). Before that, a panel on incomplete data established by the Committee on National Statistics (CNSTAT) in 1977 provided recommendations on dealing with nonresponse and published three volumes of case studies and research articles (Madow et al. 1983).

There are two main types of nonresponse error: unit nonresponse and item nonresponse. Unit nonresponse is a complete failure to obtain data from a sample unit, whether the sample unit is a household, person within a household, or a business establishment. Despite the best efforts during data collection, some level of unit nonresponse is likely to occur. Weighting techniques are often used to minimize the effect of unit nonresponse error. However, to the extent that the underlying assumptions are not fully met, such as, for example, the sample units are not missing at random, nonresponse error may still affect estimates derived from the data.

Unit nonresponse has received a great deal of attention, due in part to the frequent use of the survey response rate as an overall indicator of the quality of the data. In fact, it is often the only quantitative indicator available or widely used. Nonetheless, the nonresponse rate is only an indicator of the *potential* bias in a survey estimate due to nonresponse errors in the survey. The actual degree of nonresponse bias is a function of not only the nonresponse rate but also how much the respondents and nonrespondents differ on the survey variable of interest. Thus, the effects of nonresponse errors are very rarely directly observed due to the difficulty in obtaining information from and about the nonrespondents. Nonetheless, there are methods and special research studies that can be done to provide some information about the degree nonresponse error may affect an estimate.

Because the amount of nonresponse error is difficult to measure, efforts are often directed to minimize its occurrence. As a result, measures of the processes followed in data collection to minimize nonresponse may also serve as indicators of data quality.

Item nonresponse occurs when a responding unit does not complete an item or items on the survey questionnaire, or the response(s) obtained are unusable. Item nonresponse also affects the quality of the data. Recontact and followup efforts may be used to improve item response rates, particularly on critical data items. Measures of these efforts may also serve as indicators of data quality. Remaining item nonresponse may be dealt with by using imputation methods to compensate for the item nonresponse. These methods do not wholly compensate for the missing data, resulting in some unknown level of error remaining in the data due to nonresponse.

It is important to note that unit and item nonresponse are not completely distinct. If a survey questionnaire has been returned with responses on only some items, but not the critical items, it may be treated as a unit nonresponse. Furthermore, a mail survey questionnaire returned with no items answered may be treated as unit nonresponse, even though the lack of information on the questionnaire makes it impossible to determine whether the sample unit is in- or out-of-scope.

This chapter contains a discussion on reporting nonresponse error, including the indicators of quality pertaining to nonresponse. For unit nonresponse and item nonresponse we discuss separately the calculation of response rates, studies of nonresponse bias, and methods to compensate for nonresponse. We also discuss the processes and procedures used to minimize both unit and item nonresponse. The chapter concludes with a discussion of the reporting of nonresponse in federal surveys.

4.2 Unit Nonresponse

A variety of reasons exist for unit nonresponse, and they may vary depending on the mode of the survey and the survey design. Unit nonresponse in an interviewer-administered personal visit household survey can occur because no one is home (noncontact), refusal to participate, or inability to participate due to language barriers or cognitive or physical incapacity to respond. Sample persons in a telephone survey may refuse to participate directly, or may be using answering machines and caller ID to screen calls, and thus, in some sense refuse to participate without being formally solicited. Sample persons in mail surveys may refuse by failing to return the survey form. In this situation it can be difficult to distinguish noncontacts or nonreceipt of a survey questionnaire from refusals. Similarly, in establishment surveys, the survey questionnaire may never reach the appropriate contact person or the survey form may be sent to the wrong location, resulting in unit nonresponse.

It is important to identify and measure the different reasons and components of nonresponse because different levels of nonresponse bias may be associated with different reasons for nonresponse. For example, noncontacts, those who were never given the opportunity to choose whether or not to participate in the survey, may have very different characteristics than refusals, who were contacted but chose not to participate, and both may differ from survey respondents on some survey variables. Very different trends over time may exist for some of these components, and these trends should be monitored. For example, the increasing use of answering machines, caller ID, and call blocking devices may result in more noncontacts in surveys that use telephone and personal visits, but may appear only as cases with undetermined eligibility in surveys that use only telephone.

To define unit nonresponse for a survey it is first necessary to define the unit of analysis of the survey. Some survey designs are hierarchical and several possible levels or stages of nonresponse are possible before reaching the ultimate sampling unit. For example, an education

survey that samples classroom teachers may first sample school districts, then schools, and finally teachers. Nonresponse may occur at the district level or the school or individual teacher levels. Similarly, hierarchical designs occur in health surveys where nursing homes or hospitals are first sampled, and then in the second stage of sampling patients are sampled. Nonresponse may occur at both stages of sampling. Because of the cumulative effects of nonresponse in hierarchical surveys, even rather high levels of response at each level can still result in relatively low response rates at the ultimate sampling unit level.

Longitudinal survey designs commonly have panel nonresponse, which is unit nonresponse to one or more waves of the survey. In long-term longitudinal surveys, panel nonresponse is often permanent and results in attrition of cases from the sample, i.e., very few cases drop out one time and return the next. In this case, the effective level of nonresponse increases over time.

4.2.1 Computing and Reporting Response/Nonresponse Rates¹

As noted earlier, the response rate or its complement, the nonresponse rate, is commonly used as an indirect measure of the quality of survey data. The computation of response rates should be straightforward: the number of responding units divided by the total number of eligible units. However, there are many different ways of calculating response and nonresponse rates, and efforts to standardize these (Council of American Survey Research Organizations 1982; American Association for Public Opinion Research 2000) have not met with complete success. American Association for Public Opinion Research (2000) does, however, try to provide a general framework for response rates that can be applied across different survey modes. Smith (forthcoming) furthers this discussion by examining the development of definitions and standards concerning nonresponse rate calculations.

One difficulty inherent in response rate calculations is that the eligibility of a nonresponding case may sometimes be difficult to determine. For example, it can be very difficult to discern whether telephone numbers that are never answered are nonworking numbers versus households with no one at home. Similarly, mail surveys not returned by the postmaster are assumed to have reached the correct household or business establishment, but they may not have been received.

4.2.2 Unweighted Response Rates

As stated above, the basic definition of the survey response rate is the number of interviews with reporting units divided by the number of eligible reporting units in sample. The calculation of this rate often varies depending on the treatment of partial interviews and the treatment of sample units of unknown eligibility. The discussion that follows has benefited from the work of a committee of the American Association for Public Opinion Research (2000).

To illustrate the calculation of some common response rates, it is necessary to define the possible outcomes. For the purpose of this discussion, all partial interviews have been reclassified as a completed interview or a noninterview depending on the extent to which the questionnaire was completed. With this simplification, survey cases can be categorized into four groups: interviewed cases, eligible cases that are noninterviews, cases of unknown eligibility, and cases that are not eligible.

¹ This section draws heavily on work done by a previous FCSM subcommittee on nonresponse.

We let:

- I = number of interviewed cases
- R = number of refused interview cases
- NC = number of noncontact sample units known to be eligible
- O = number of eligible sample units, noninterviewed for other reasons
- U = number of sample units of unknown eligibility, no interview
- e = estimated proportion of sample unit cases of unknown eligibility that are eligible

The most useful and simple response rate for surveys in which sample units have an equal probability of selection and the frame includes no ineligible units is:

$$RR1 = \frac{I}{I + R + O + NC} \quad (4.1)$$

This response rate is the one most often used in interviewer-administered federal household surveys. For example, the Current Population Survey (CPS), the National Crime Victimization Survey (NCVS), the Consumer Expenditures (CE) Survey, and the National Health Interview Survey (NHIS) conducted by the U.S. Bureau of the Census all report response rates using this formula.²

When the sample frame is imperfect and some cases have uncertain eligibility, as is common in random digit-dialing (RDD) telephone surveys in which a phone is not answered after repeated attempts, the denominator should be an estimate of the total number of eligible units. Thus, one may compute a response rate using equation 4.1, ignoring the cases, U , with undetermined eligibility—and overestimate the true response rate—or include those cases in the denominator—and probably underestimate the true response rate. In this case, surveys may report more than one response rate with the true value somewhere in between. Conversely, they may attempt to estimate the true rate by estimating the number of the cases that would be eligible using an estimate of the probability of a case being eligible given that it is undetermined.

If we let e be the estimated proportion of cases of unknown eligibility that are eligible, then, e multiplied by the number of undetermined cases, U , as shown in equation 4.2 is the estimated number of sample unit cases eligible for interview from the sample unit cases of unknown eligibility. The proportion, e , may be based on results from the current sample or may be based on other research or surveys or guidelines such as those put out by the Council of American Survey Research Organizations (CASRO).

$$RR2 = \frac{I}{I + R + O + NC + eU} \quad (4.2)$$

For example, the National Household Education Survey (NHES) is an RDD survey sponsored by the National Center for Education Statistics that reports four response rates, each of which uses a different set of assumptions for the allocation of the cases with unknown residential status (table 4.1). The first response rate uses the proportion of households *identified in checks with telephone*

² The U.S. Bureau of the Census also calculates a nonresponse rate that includes the R , NC , and O cases into a category called Type A nonresponse. The Type A nonresponse rate is the complement to the response rate.

business offices to allocate the proportion of cases with unknown residential status (U) in the denominator. This rate was 40.5 percent for the study cited in table 4.1. The second response rate (*CASRO response rate*) uses the proportion of households observed in the study with known residential status ($e=47.2$ percent) to allocate the proportion of cases with unknown residential status (U) in the denominator. These rates are both based on equation 4.2. The difference between the response rates in table 4.1 is due to the estimate used for e . The other two response rates in table 4.1 are a conservative response rate, which includes all cases (U) of unknown eligibility as eligible (that is, $e=1$) and a liberal response rate, which excludes all cases of unknown eligibility (that is, $e=0$) (table 4.1).³

Table 4.1.—Example of reporting multiple response rates for random digit-dialing (RDD) surveys: 1996 National Household Education Survey

Screeners response category	Number	Percent of all numbers	Percent of residential numbers
Total	161,446	100.0	
Identified as residential	76,258	47.2	100.0
Participating	55,838	34.6	73.2
Not participating	20,420	12.6	26.8
Identified as nonresidential	75,736	46.9	
Unknown residential status	9,452	5.9	
Screeners response rates*			Rate (Percent)
Estimated response rate (using business office method)			69.9
CASRO response rate (using known residential data)			69.1
Conservative response rate (using all U , $e=1$)			65.4
Liberal response rate (using no U , $e=0$)			73.2

* All of the response rates use the weighted number of participating households as the numerator (see table 1 from Montaquila and Brick 1997, 17).

SOURCE: Montaquila, J. and Brick, J. M. 1997. *Unit and Item Response Rates, Weighting, and Imputation Procedures in the 1996 National Household Education Survey*. Washington, DC: U.S. Department of Education. National Center for Education Statistics (Working Paper No. 97–40).

When the sample design is hierarchical, nonresponse can occur at any stage, e.g., a school survey that selects schools and teachers within schools. In this case, one can estimate the proportion of eligible units that were not selected because of nonresponse at a higher level using equation 4.2.

³ The NHES publications and technical reports typically report all of these as weighted response rates, which are described below; however they are also calculated as unweighted response rates to monitor data collection.

Alternatively, one can compute response rates at each level using equation 4.1 and then calculate the final response rate by multiplying the rates at each level. For example, the response rates for the NCES' Teacher Survey from the Schools and Staffing Survey (SASS) are calculated and reported in this manner (table 4.2).⁴

Table 4.2.—Hierarchical response rates: Response rates for the Schools and Staffing Surveys, public schools 1993–94

Public school component	Sample size	Unweighted response rate	Weighted ¹ response rate	Weighted overall ² response rate
Local Education Agency (School District) Survey	5,363	93.1	93.9	93.9
Administrator	9,415	96.6	96.6	96.6
School	9,532	92.0	92.3	92.3
Teacher	53,003	88.9	88.2	83.8
Library	4,655	91.1	90.1	90.1
Librarian	4,175	93.5	92.3	92.3
Student	5,577	90.2	91.3	80.3

¹ Weighted using the inverse of the probability of selection.

² The weighted overall teacher response rate is the product of the weighted response rate of teachers and the response rate of schools providing a list of teacher (95 percent). The weighted overall student response rate is the product of the weighted student response rate and the response rate of schools providing a list of students (88 percent).

SOURCE: Excerpted from Monaco, D., Salvucci, S., Zhang, F., Hu, M., and Gruber, K. 1998. *An Analysis of Total Nonresponse in the 1993–94 Schools and Staffing Survey (SASS)*. Washington, DC: U.S. Department of Education, National Center for Education Statistics (NCES 98–243).

4.2.3 Weighted Response Rates

It is common for household and establishment surveys to have different probabilities associated with different units in the sample in order to reduce the variance in subpopulation estimates or estimates for the total population. In this case, it is useful for the response rate to reflect the selection weight, w_i . For each observation i , let $I_i = 1$ if the i th respondent is an interview, and $I_i = 0$, if the i th respondent is not an interview. Similarly, let $R_i = 1$ if the i th sample unit is a refusal, and $R_i = 0$ if the i th sample unit is not a refusal; $NC_i = 1$ if the i th sample unit is a noncontact sample unit known to be eligible, and $NC_i = 0$ if the i th sample unit is not a noncontact sample unit known to be eligible; and $O_i = 1$ if the i th sample unit is a noninterview for reasons other than a refusal and known to be eligible, and $O_i = 0$ if the i th sample unit is a noninterview for reasons other than a refusal and known to be eligible.

⁴ The SASS Teacher Survey response rates are also calculated both weighted and unweighted.

A weighted response rate, with w_i = the inverse of the probability of selection for the i th sample unit and the sum is over all sample units selected to be in sample, can be given by:

$$\frac{\sum w_i I_i}{\sum w_i (I_i + R_i + NC_i + O_i)} \quad (4.3)$$

Equation 4.3 can be modified to reflect cases of unknown eligibility. This response rate shows the estimate of the proportion of the population measured if the same survey procedures were used on a census of the population. This rate may be useful for characterizing how completely the population was measured prior to adjustment for nonresponse and can also be useful for showing separate response rates for subpopulations that had different selection probabilities.

In addition, many establishment surveys use weighted response rates because a small number of extremely large firms may dominate an industry (Osmint, McMahon, and Ware-Martin 1994). Because nonresponse by one of these large firms can have severe consequences on how well a variable, such as total sales, is estimated, weighted response rates in establishment surveys are typically reported as an estimate of the proportion of the population total for a characteristic, y , associated with the responding sample unit. The characteristic, y , for example might be “total assets” or “total revenues” or the “total amount of coal produced.” This response rate is often called a “coverage rate” but, in fact, is a weighted item response rate, where the item of interest is a quantity of primary interest for the survey. If we let y_i be the value of the characteristic y for the i th sample unit and sum over the entire sample, then the weighted response rate is given as:

$$\frac{\sum w_i y_i I_i}{\sum w_i y_i (I_i + R_i + NC_i + O_i)} \quad (4.4)$$

Equation 4.4 can be modified if there are cases of unknown eligibility. Data collected for a previous period or from administrative records may be used in the denominator in a weighted response rate to represent the nonrespondents.

4.2.4 Using Weighted versus Unweighted Response Rates

Both unweighted and weighted response rates can be useful for different reasons in any survey. Unweighted response rates provide an indicator of the quality of the data collection and may be calculated at national, regional, and interviewer levels or for certain domains to evaluate performance. The unweighted nonresponse rate (or the number of nonrespondents) indicates the extent of nonresponse followup—a data collection workload measure. Weighted response rates indicate the proportion of the population (or some calculated subpopulation) that responded to the survey, and can be useful for an analyst’s evaluation of the effect of nonresponse on survey estimates (Kasprzyk and Kalton 1997; Madow, Nisselson, and Olkin 1983). For example, the SASS and NHES surveys (both sponsored by NCES) rely on unweighted response rates during data collection to monitor progress. However, they analyze weighted nonresponse rates to determine whether portions of the population are underrepresented in their surveys due to nonresponse (Jabine 1994; Scheuren et al. 1996). For establishment surveys, the weighted response weight reflects the proportion of some characteristic of interest covered by the sample, and reflects the amount the characteristic will need to be estimated by using imputations or weight adjustments.

4.2.5 Other Response Rates

Longitudinal surveys also present some special concerns for the calculation of response rates. Longitudinal surveys are not only subject to nonresponse at the initial interview as a result of refusals, unable to participate, etc., but also are subject to nonresponse for the same reasons at later interviews. If nonrespondents in one wave of a longitudinal survey are eligible sample units in subsequent waves of a longitudinal survey, response rates corresponding to the missing interview pattern can be calculated (Lepkowski 1989). The response rate in longitudinal surveys, however, often focuses on attrition, i.e., those who are not interviewed again after being interviewed one or more times. For surveys that are person-based, it is common to report a retention rate, which indicates the percentage of the persons interviewed who were interviewed at later waves divided by those eligible for interview. For example, the National Longitudinal Survey (NLS) reports retention rates for each year's interviewing. In this case, it may be difficult to determine the eligibility of some sample persons who are not locatable.

Longitudinal surveys that are household- and person-based have further complications because households may leave the survey universe or split up after the first interview, and new persons may also enter established households. All of these changes require that more persons and households be located and tracked over time. Thus, the exact number of eligible households or persons may not be known from nonresponding or unlocatable households. For example, in the Survey of Income and Program Participation (SIPP), Lepkowski, Kalton, and Kasprzyk (1989) estimated the number of nonresponding persons from the number of nonrespondent households by multiplying the average number of sample persons in responding households by the number of nonresponding households.

It may also be useful to calculate other response rates for specific purposes. For example, in a mail survey one may want to report a completion rate—the ratio of completed questionnaires to all questionnaires mailed out. This rate would be the same as equation 4.1 except that one would include the cases with undetermined eligibility (*U*) as well as ineligible or out-of-scope cases in the denominator. In addition, economic surveys that release preliminary estimates before all data are collected may provide response rates for the preliminary and final stages and any others in between.

It may be useful to distinguish between refusals and noncontacts to identify different kinds of problems in field procedures. One may compute a contact rate to see how many households are not being reached, as well as a cooperation rate, which reflects the percentage of contacted cases that actually participate. A high noncontact rate may indicate lack of interviewer effort, or poor timing, while a low cooperation rate may reflect the need for better training in selling the survey or stronger efforts on refusal conversion.

It is also common to calculate response rates at various levels to monitor data collection and performance of interviewers. For example, the U.S. Bureau of the Census typically calculates response rates for each regional office for many of its surveys. In turn, the regional offices calculate response rates for each interviewer or team of interviewers for particular surveys. Most data collection organizations include response rates as a component (sometimes the major component) in their evaluation of interviewers.

Finally, surveys that use different modes of data collection, such as initial mailings followed by telephone interviews with nonrespondents, with final followup by personal visit for difficult or reluctant respondents, may want to calculate the response rates for each mode as well as an

overall response rate. This allows for the analysis of data quality associated with individual modes of collection. For example, in the Current Population Survey (CPS), the U.S. Bureau of the Census calculates a recycle rate for cases that are sent to a centralized telephone center for interviewing. Cases that are not contacted by the telephone center or refuse to participate are sent back (or “recycled”) to the field interviewer for followup. A response rate for field interviewer followup cases would be informative as well.

4.2.6 Measuring and Reporting Nonresponse Bias

As noted earlier, the degree of nonresponse error or nonresponse bias is a function of not only the nonresponse rate but also how much the respondents and nonrespondents differ on the survey variables of interest. For a sample mean, an estimate of the bias of the sample respondent mean is given by:

$$B(\bar{y}_r) = \bar{y}_r - \bar{y}_t = \left(\frac{n_{nr}}{n} \right) (\bar{y}_r - \bar{y}_{nr}) \quad (4.5)$$

where:

$\bar{y}_t$ = the mean based on all sample cases

$\bar{y}_r$ = the mean based only on respondent cases

$\bar{y}_{nr}$ = the mean based only on nonrespondent cases

n = the number of cases in the sample

n_{nr} = the number of nonrespondent cases

$\bar{y}_r$ is approximately unbiased if either the proportion of nonrespondents (n_{nr}/n) is small or the nonrespondent mean, $\bar{y}_{nr}$, is close to the respondent mean, $\bar{y}_r$.

Nonresponse may also result in an increase in the mean square errors of survey estimates, a distortion of the univariate and multivariate distributions of survey variables, and therefore result in biased estimates of means, variances, and covariances. Because of the smaller sample size due to nonresponse, variances of estimators are also increased.

The nature of nonresponse is such that values for nonrespondents on all survey measures are not available. A variety of methodologies have been developed and used to estimate the amount of nonresponse bias for at least some survey variables. Some of these methodologies require conducting special studies, while others take advantage of information available in the sampling frame or information gathered during data collection.

The potential for nonresponse bias can be assessed in what are often called identification studies (e.g., see Lessler and Kalsbeek 1992). In this type of study the characteristics of respondents and nonrespondents are compared on a variety of sociodemographic characteristics available from the sampling frame or some external source. If the distribution of the available variables is similar for respondents and nonrespondents, then the concern over nonresponse bias is lessened. In contrast, if the distributions are different the observed differences on the available variables may provide some insight as to the differences that exist on other survey measures. For example, analysis of the school nonresponse rates in the 1988 National Education Longitudinal Study found differences by type of school (public/private) and region of the country, but not by school

size or minority enrollment. Furthermore, analysis of 14 key items collected from both responding and nonresponding schools only identified potential bias in three of the key items, all of which were related to low response rates in public schools (Spencer et al. 1990).

One variation of an identification study is to examine the characteristics of respondents to different waves of data collection, most commonly in mail surveys. In a study of respondents to the 1993 survey of Doctorate recipients, Moonesinghe, Mitchell, and Pasquini (1995), examined the weighted responses of respondents who were “early” (responded to the first wave mailing), “interim” (responded to the second or first mailing), and “late” (interim plus CATI followup interviews). They found that those who were academically employed, teaching, and full professors may be under-represented among the “early” respondents, as compared to the “interim” or “late” estimates, thus potentially biasing the “early” estimates.

A variation of this approach can be found in Tucker and Harris-Kojetin (1998) who take advantage of panel survey data from the Current Population Survey. They use information from the previous month in which the household reported to understand the characteristics of nonrespondents and the consequence of nonresponse on labor force estimates. They compare characteristics of households that are respondents in two consecutive months with those that are nonrespondents in one of the two months.

Another variation of an identification study is to match a sample of survey respondents and nonrespondents with a complete data source, such as a census. For example, Groves and Couper (1998) report results from six federal household interviewer-administered surveys that they matched with 1990 decennial census data to learn about the demographic characteristics of those who did not participate in large-scale, face-to-face interviews. Using household and block-level information from the census, they developed distinct multivariate models for households that were contacted compared to those not contacted, and households that cooperated compared to those that were contacted, but did not cooperate. In the case of contact rates, availability and access seem to play a role. Single family homes, households with young children, and the elderly tend to have higher contact rates; while households in multiunit structures, single person households, and households in urban areas tend to have lower contact rates. Cooperation rates tend to be higher in older households, households with young children, and households with young adults; but those who live alone tend not to cooperate.

The 1987–88 National Food Consumption Survey (NFCS) compares demographic characteristics of respondents with estimates from the Current Population Survey to establish indirect inferences regarding nonresponse bias that may result if the unweighted NFCS data were used for analysis (Guenther and Tippet 1993). These analyses are part of a larger body of work described in Guenther and Tippet (1993) in which comparisons on other contemporaneous surveys and past surveys are undertaken to establish the extent of bias due to nonresponse. In addition, because the NFCS consisted of several survey components, intrasurvey comparisons were made on the same items across components to look for potential patterns of nonresponse bias.

Random digit dial (RDD) surveys typically have little or no frame information, making it difficult to compare respondents and nonrespondents. However, the National Household Education Survey (NHES) mapped their RDD telephone exchanges onto zip codes and used census information about households in those zip codes to compare areas with higher and lower response rates and, thus, obtain some indication of possible nonresponse bias (see Montaquila and Brick 1997).

Another methodology that can be used is to subsample nonrespondents to a survey and use extensive followup and conversion procedures to obtain complete cooperation from that subsample. Survey statistics can then be calculated for the subsampled nonrespondents and used to estimate parameters for all of the nonrespondents. The National Education Longitudinal Survey (NELS:88) used these techniques between the base year and the first followup to improve the estimates for students with disabilities and for students with limited English proficiency (Ingels 1996).

4.3 Item Nonresponse

As noted earlier, item nonresponse occurs when a respondent provides some, but not all, of the requested information, or if the reported information is not useable. The line between item and unit nonresponse is sometimes not clear. If a respondent breaks off an interview or sends in a partially completed questionnaire, at what point is it considered item nonresponse versus unit nonresponse? To the extent that survey managers draw the line at different points, there can be differences in the level of unit nonresponse as the level of item nonresponse that is tolerated changes. For example, if a particular survey manager requires 90 percent of all items to be answered for a completed questionnaire, it is possible that a number of partial interviews would be treated as unit nonresponse. But if a different survey manager changed the required level of item response to 80 percent, the number of partial interviews treated as unit nonresponse would decrease and the unit response rate would increase.

4.3.1 Causes of Item Nonresponse

Item nonresponse may occur for a variety of reasons and in a variety of ways (Groves 1989; Madow, Nisselson, and Olkin 1983). A number of items may be missing because the respondent broke off the interview after partially completing it, but enough data were provided so that the questionnaire is not classified as a unit nonresponse. Single or multiple items may be missing because the respondent inadvertently skips an item or block of items on a self-administered questionnaire or refuses to answer the question(s). Sometimes, the respondent may not have the information to answer the question, and this may occur more frequently when the respondent is a proxy for another person. However, Groves (1989) also notes that “don’t know” is a meaningful response for some questions such as those inquiring about opinions of a political candidate or a referendum. Finally, items may be missing because the interviewer skips an item or block of items or fails to record a respondent’s answer.

The greatest concern with the respondent’s refusal to provide the requested data is that there may be systematic differences between those who provide the information and those who do not. Respondents may have good reasons for refusing to answer certain items. Item-missing rates are frequently highest for issues people perceive as private or sensitive, such as income. Items that require greater cognitive effort or confuse the respondent may also have higher nonresponse rates. Because item-missing data occur after the respondent agrees to participate, item nonresponse rates can also be important quality indicators that reflect problems in survey question wording or response options. In other words, item-missing data also reflect measurement errors as well as nonresponse errors (Groves 1989), and many of the issues discussed in chapter 6 are quite relevant to item nonresponse. For example, the quality profile of the SASS (Jabine 1994) describes research relating problems in the design and layout of questions to high item nonresponse rates. Changes were made to some items and the differences

in nonresponse rates between rounds 1 and 2 of the survey showed considerable improvement in item nonresponse due to these design changes.

Nonresponse rates for specific items or sections of a questionnaire are an important quality indicator for data items in the same way that the unit nonresponse rates are used as overall indicators of the quality of survey data. Furthermore, agencies may set requirements at certain levels of overall response prior to developing the data for analysis, publication, and public use datasets.

4.3.2 Computing and Reporting Item Response/Nonresponse Rates

Item nonresponse rates are typically computed only for responding units. The item nonresponse rate is usually calculated as the ratio of the number of eligible units responding to an item to the number of responding units eligible to have responded to the item (Madow, Nisselson, and Olkin 1983). It may be useful to calculate item nonresponse rates for specific important items in a survey or for blocks of questions in the questionnaire or an overall level for the entire questionnaire. For example, the quality profile for the American Housing Survey shows the item nonresponse rates for 43 selected items (Chakrabarty and Torres 1996).

It is also useful to report item nonresponse rates separately for different kinds of respondents or by important survey variables. For example, the National Science Foundation's Survey of Graduate Students and Postdoctorates in Science and Engineering reports item-missing rates⁵ separately for different subject areas, for example, Computer Sciences, Biological Sciences, as well as for full-time and part-time graduate students and postdoctoral students (National Science Foundation 1999).

As with unit nonresponse rates, item nonresponse rates can be computed at different levels. It can be useful to calculate item-missing rates for interviewers or groups of interviewers to monitor the quality of the data they are collecting. In a household survey, it may be useful to compute item-missing rates separately for the whole household as well as for individuals within the household.

4.4 Compensating for Nonresponse

Given that a survey has encountered some level of nonresponse, the survey manager is faced with the problem of how to compensate for it. Although there are several options including doing nothing and working harder to decrease the number of nonrespondents (see Lessler and Kalsbeek 1992, for several alternatives), typically some method is used to attempt to compensate for nonresponse. There are two general approaches to compensating for nonresponse, adjustment as part of the estimation process (e.g., weighting adjustments) and directly estimating the value each nonrespondent might have reported (imputation) had he/she been a respondent. Kalton and Kasprzyk (1986) note that weighting adjustments are primarily used to compensate for unit nonresponse while imputation procedures are more likely to be used to compensate for missing items. For some periodic surveys, such as those conducted by the Energy Information Administration, imputation procedures are used to estimate all survey items for unit respondents.

⁵ The item-missing rates are reported as imputation rates, because missing items are imputed, see section 4.2.3.

4.4.1 Weighting Procedures

A variety of different weighting procedures and models can be used, e.g., population weighting, sample weighting, ratio, and response propensity. See, for example, Kalton and Kasprzyk (1986), Lessler and Kalsbeek (1992), Oh and Scheuren (1983), Brick and Kalton (1996) for summaries of different procedures. The purpose of all these procedures is essentially to increase the weights of the respondent cases to represent the nonrespondents.

Nonresponse weighting adjustments require some information about the nonrespondents and respondents as well as assumptions about the differences between respondents and nonrespondents. Except for special studies (as described in the previous section) little is known about nonrespondent cases other than information that is contained on the sampling frame. For example, in the Current Population Survey, the nonresponse adjustment is based only on clusters of similar sample areas (using Metropolitan Statistical Area (MSA) status and size) that usually are contained within a state. Each MSA cluster is further split into “central city” and “balance of MSA,” while non-MSA clusters are split by “urban” and “rural” residence.

In the case of panel studies, data from the initial interview can be used to develop nonresponse adjustments to the weights for future waves of interviews. Reporting on the information used to adjust for nonresponse can be valuable for analysts who wish to study alternative methods of adjustment or evaluate the utility of the adjustment procedures. Lepkowski, Kalton, and Kasprzyk (1989) used longitudinal data from the SIPP to examine the characteristics of panel nonrespondents and evaluate alternative nonresponse weighting adjustments. They found that the majority of nonrespondents were interviewed at least once, and thus a great deal was known about them from the interview(s) they gave. After finding significant differences between complete and partial respondents on a variety of demographic characteristics, they used alternative modeling techniques to develop new weight adjustments. They compared and evaluated the effects of the different weights with the initial panel weights by examining how highly correlated they are with each other and by comparing survey estimates produced with each set of weights. As a final step, they compared weighting procedures by examining the distribution of weights that they produce. The more variable the weights are, the greater the loss of precision in the survey estimates. Overall, their analyses did not show the alternative weighting schemes to be more effective in compensating for panel nonresponse than the method being used.

Harris-Kojetin and Robison (1998) used panel data from the Current Population Survey to examine the effects of unit nonresponse. Since most CPS nonrespondents are actually respondents for at least one month when in sample, data obtained from other months can be used both to compare respondents and nonrespondents as well as to evaluate the unit nonresponse adjustment procedures.

Because weighting for nonresponse can affect the quality of the estimates from a survey, it is important for agencies to research the effects of nonresponse adjustments on statistical estimates. To evaluate nonresponse adjustment procedures typically requires some external source of data similar to that used in nonresponse bias studies. For example, a National Center for Health Statistics study revealed that a weighting procedure used during the 1988 cycle of the National Survey of Family Growth may have reduced nonresponse bias. The researchers compared estimates of the number of United States births resulting from the 1988 weighting procedure with

administrative records and found the procedure produced more accurate estimates than the 1982 weighting procedure (see Mosher, Judkins, and Goksel 1989).

As noted in section 4.1, it may be important to distinguish between refusals, noncontacts, and other noninterviews because the underlying causes of each type of nonresponse may be quite different. Nonresponse adjustment procedures have typically not taken into account the different kinds of nonresponse, but, in an extensive study, Groves and Couper (1998) match sample survey data to the 1990 U.S. Decennial Census to create separate models of cooperation and contact from six federal surveys. They suggest how these models could be used to improve nonresponse adjustment procedures. Given the differences they found between models for noncontact and refusals, they suggest that the source of nonresponse and the related correlates should be taken into account when response propensity models are used to develop nonresponse weight adjustments.

Nonresponse adjustment models can only incorporate information obtained on both respondents and nonrespondents. This has resulted in recommendations that further efforts be devoted to obtaining additional information on nonrespondents as part of the survey design to enrich and improve the nonresponse adjustment models (Groves and Couper 1995; Madow, Nisselson, and Olkin et al. 1983).

4.4.2 Imputation Procedures

Typically, a great deal more data are available to examine the impact of bias due to item nonresponse than unit nonresponse because respondents have answered a sufficient number of items in the survey to be considered at least a partial interview. As a result, comparisons can be made between respondents and nonrespondents on the items that were completed to attempt to infer whether there are any systematic differences. Frequently, the data from the items that were answered are used to model or impute values for those that are missing.

Survey managers must decide how to deal with item-missing data. One could do nothing and conduct analyses of data on only complete cases. This results in the loss of some usable data and a reduced sample size. However, using only complete cases is not an option if the primary focus of the survey is to provide aggregates such as total sales or total revenues. Alternatively, one could use a similar strategy to unit nonresponse and weight the respondents to adjust for the item nonresponse. However, this could require a different weight for each survey item with missing data. Thus, it is most common to impute for item-missing data rather than weight (Kalton and Kasprzyk 1982).

There are a wide variety of procedures used to impute for item missing data including hot-deck methods, regression methods, and mean imputation (For more information, see Brick and Kalton 1996; Kalton and Kasprzyk 1982 and 1986; Lessler and Kalsbeek 1992; Little and Rubin 1987.). Although imputation methods yield data values for missing items, they come at a cost. Imputation may distort measures of the relationships between variables, and they also distort standard error estimation. There are advantages and disadvantages to each of the different imputation methods. Multiple imputation (Rubin 1987 and 1996) or the assigning of more than one imputed value to a missing data item has been discussed widely during the last 20 years. A major advantage of multiple imputation (Herzog and Rubin 1983; Little and Rubin 1987) methods is that they allow estimates of standard errors to be computed by using the same imputation procedure several times. A disadvantage is that analysts are not accustomed to using

multiple imputations in their analyses. The National Center for Health Statistics has developed a data set (available Summer 2001) for the National Health and Nutrition Examination Survey (NHANES) that includes five imputations with detailed instructions for using the data.

Because each imputation method relies on some model (implicit if not explicit) for generating values for the missing items, it is useful to report the method of imputation and variables used in the imputation model, or to identify donor records, or define imputation classes. For example, the American Housing Survey uses prior year data for some items and uses a sequential “hot deck” procedure for other items. The variables used to define the imputation cells vary, but to impute income they use age, race and sex of the person, relationship to reference person, and value of property/monthly rent (Chakrabarty and Torres 1996).

In addition, indicator or shadow variables for each variable that was imputed can be valuable for analysts who wish to use some other form of imputation or analyze only complete records (see section 6.2.4, for example). Indicators of how often a record was used as a “donor” in hot deck imputation can also serve as a quality indicator of the imputation process.

Because imputation for item nonresponse can affect the quality of the estimates from a survey, it is important for agencies to research the effects of those methods. There are several basic methodologies for evaluating the effect of imputation on the data. One way of determining the effect of imputation is to compare estimates that include the imputed values with estimates based only on reported data. Another way of determining the effect of imputation is to compare the estimates to external sources. For example, the CPS hot deck imputation procedure for income was evaluated by using data from the March 1981 CPS income supplement matched to 1980 Internal Revenue Service (IRS) tax records (David et al. 1986).

Agencies can provide in their documentation and in their public-use datasets a great deal of information useful to analysts in evaluating the effects of imputation. For example, the methodology report for the 1992–93 National Postsecondary Student Aid Study (Loft et al. 1995) includes a technical appendix that describes the imputation procedures used for variables that required imputation. Also included are comparisons of the pre- and post-imputation values for the variables.

4.5 Methods and Procedures to Minimize Unit and Item Nonresponse

Federal statistical agencies have used a variety of methods to minimize both unit and item nonresponse, with the goal of improving data quality. The extent of activity to minimize nonresponse can also be an indicator of data quality, and descriptions of these efforts can provide the data user with useful information. Most recurring publications mention that there is followup for nonresponse, but typically do not contain a detailed discussion of how unit or item nonresponse is minimized. Technical and methodological reports provide more detail. An example is the U.S. Energy Information Administration’s *Quality Profile of the Residential Energy Consumption Survey* (U.S. Energy Information Administration 1996). The quality profile describes the use of monetary incentives to improve completion rates of automobile use diaries.

There is extensive literature on minimizing nonresponse. See, for example, the report concerning establishment surveys from the Federal Committee on Statistical Methodology (1988) and more

recently, Paxson, Dillman, and Tarnai (1995). A number of examples for household surveys have also been presented (Dillman 1978; Groves 1989; Lyberg and Dean 1992). The methods can be grouped into three categories: front-end techniques to promote cooperation, special arrangements, and followup techniques. Recently, some agencies have considered shifting their focus from followup to front-end techniques.

Front-end Techniques. There is no empirical evidence that a single technique produces high response rates. Further, different surveys require different techniques. Studies have shown that positive effects are obtained when using a combination of techniques for some surveys.

Techniques that apply to surveys conducted by interviewers and that are self-administered (both household and establishment surveys) include advance notification in the form of a letter or phone call, use of priority mail, personalization of correspondence, and showing respondents how the data they are providing are being used. Techniques specific to surveys conducted by interviewers include performance guidelines for interviewers and the monitoring and observation of interviewer performance. Techniques specific to self-administered surveys are the use of respondent-friendly questionnaires, that is, questionnaires that are designed to appear easier and less time-consuming to complete, and the use of stamped-return envelopes. Incentives, both financial and nonfinancial, may also help increase response rates. Many in the survey research community believe incentives are an important tool to improve response rates, particularly with a difficult-to-interview population. Kulka (1995) provides a review of the research literature on incentives and a description of current practices.

Special Reporting Arrangements. An example of a special reporting arrangement for small companies is the use of sample rotation so that companies do not have to report each time the survey is conducted, or the use of a shorter version of the form by special arrangement.

Followup Techniques. The Federal Committee on Statistical Methodology (1988) reports that over three-fourths of agency surveys included in their study use intensive followup of critical units and nearly all use some type of nonresponse followup procedure. Followup techniques include reminder cards, periodic telephone calls, and automated fax reminders. Followup techniques help increase the overall response rate; however, if left unmanaged, production schedules can be delayed and the timeliness of the data products affected. Dillman (1991) recommends an integrated approach for front-end and followup techniques so that they are not duplicative.

4.6 Quality Indicators for Nonresponse

Nonresponse errors are likely to be present to some degree in any survey, and they may affect the quality of the data and estimates produced from the survey. The exact amount of nonresponse error on a specific estimate is almost never known. However, there are a number of indicators of the quality of those data that can be extremely useful to analysts, customers, and consumers in evaluating the quality of the data and their usefulness.

Frequently the response rate (or its complement, the unit nonresponse rate) is used as an overall indicator of the quality of the data in the survey. There are also a variety of other more specific indicators that provide insights into the quality of the data and the data collection process. Specifically, rates of refusals and noncontacts can be useful when computed for the entire data collection as well as at the level of field administrative units or interviewers. Also, address not

locatable, postmaster returns, and undetermined eligibility (e.g., answering machine—eligibility unknown) can be important in monitoring and evaluating mail and telephone surveys.

Some similar quality indicators can be calculated for item nonresponse rates. These provide a more micro-level data quality indicator for a specific estimate or set of estimates, but also may be informative about the data collection process when computed at different levels.

Additional indicators can be useful in evaluating the potential for nonresponse error in a survey by focusing even more on the data collection process itself. Documenting the procedures that were followed to minimize nonresponse during the data collection may provide a user with helpful information to evaluate the likelihood of nonresponse errors. For example, quality indicators may include the number of contacts attempted for noncontact cases, the attempts and techniques used to convert refusals, and the number of and kinds of reminders and replacement questionnaires sent for mail surveys. One may have more confidence in the results from a survey that had a lower than desired response rate if systematic and extensive procedures were followed to minimize nonresponse.

Indicators of the amount of nonresponse bias for some estimates may be found through special studies of nonrespondents or by using external data. Furthermore, because steps are often taken in the estimation process to compensate for nonresponse by weighting and imputation, effects of these procedures on the quality of the data can be shown by using external data or special studies of nonrespondents. For example, the validity of the weighting or imputation model used can be evaluated by making direct comparisons of the estimates, generated with and without the procedure, to a reliable external data source.

4.7 Reporting Nonresponse in Federal Surveys

Nonresponse is the most visible and well-known source of nonsampling error. Nonresponse rates are frequently reported and are often viewed as the first area requiring study to assess the potential for bias in survey estimates. As discussed above, response rates are a common output of the data collection process and are calculated differently for different purposes. These rates tend to be treated as a proxy for survey data quality—more so than almost any other indicator that may be proposed. Response rates influence the perception of the overall quality of the survey—low response rates suggesting a poor quality survey—and they are usually available.

In the subcommittee’s study of the reporting of error sources in analytic publications, Atkinson, Schwanz, and Sieber (1999) found, somewhat surprisingly, that nonresponse was not always identified as a potential error source and response rates were frequently not reported. Of the publications reviewed, 71 percent mentioned unit nonresponse as a possible source of nonsampling error and only 59 percent indicated an overall response rate. Forty-nine percent of the publications that were reviewed mentioned item nonresponse as a source of error, and only 22 percent reported item nonresponse rates. Twenty percent of the publications presented response rates for subgroups.

These findings are very surprising and somewhat disappointing to those who expect a high reporting standard of the federal statistical agencies. Since response rates are perhaps the most easily obtained measure of survey performance and are usually available routinely—without additional computational effort—precisely defined response rates should be reported routinely in analytic survey reports. As with other sources of error, certain kinds of information about

nonresponse is not likely to occur in the technical notes section of analytic reports. Studies to estimate the bias due to nonresponse—either by comparing respondents and nonrespondents characteristics on variables available on the sampling frame or by studying a subsample of nonrespondents who have had intensive followups are likely to be reported in special methodological or technical publications. Synthesis reports that bring together a substantial amount of information about a survey are available as user’s guides, survey technical documentation, and quality profiles. In each of these types of reports, substantially more material about nonresponse error can be made available to the user.

The discussion of nonresponse error goes beyond quantifying its magnitude; it also includes the precision of the definition, identification of the nonresponse components, and the nature and effects of procedures to compensate for nonresponse. The nature of reporting may depend on the magnitude of the nonresponse problem. For example, using a variable with a high nonresponse rate in a key analysis should warrant some discussion in the report, whereas, if the variable had a low nonresponse rate, no discussion may be necessary. In the course of reviewing errors due to nonresponse, the subcommittee identified a number of aspects of this source of error that need to be addressed when reporting results in an agency analytic report:

- Unit and item nonresponse should be identified as important sources of error.
- Overall unit response rates (weighted using base weights and unweighted) should be provided as well as definitions of the response rates given; this includes providing definitions of the numerator and denominator in the response rate calculation.
- In multistage designs, weighted (using base weights) and unweighted response rates at each interview level should be given, and an overall response rate computed. Assumptions necessary for the response rate calculation should be stated.
- Longitudinal surveys should report separately the response rate for the first wave of the survey, each followup wave of the survey, as well as the cumulative response rate. Reporting other response rates is encouraged; for example, the response rate of sample units responding in all waves of a longitudinal survey may be informative to the longitudinal data analyst.
- Subgroup response rates should be provided if specific subgroups are important to the analysis found in the report.
- Item response rates should be summarized and items with low response rates identified.
- Unit and item nonresponse adjustment procedures—whether they are weighting procedures or imputation methods—should be identified.
- When unit or item nonresponse rates are lower than the agency seems “reasonable,” special studies to assess the bias due to nonresponse should be conducted, the results summarized, and the detailed report referenced.
- If available, studies designed to measure potential nonresponse bias should be referenced.

Technical reports and user’s manuals should address and document the myriad important details related to nonresponse as a source of error in surveys, including:

- Procedures used to compensate for missing data, both unit and item nonresponse, should be described and the key variables used in the procedures identified. The effects of these procedures (if known) on the estimates should be discussed.
- Evaluations of the missing data procedures, both unit and item, should be conducted, the results summarized, and a detailed report referenced.
- Special studies that aim to understand or measure the bias due to nonresponse should be conducted, summarized, and the detailed report referenced.
- Steps taken to maximize the response rate and the extent of nonresponse followup should be described.
- Subgroup response rates for key subpopulations should be calculated and made available.
- Reasons for nonresponse (refusals, noncontacts, etc.) should be monitored and reported.

References

- American Association for Public Opinion Research. 2000. *Standard Definitions: Final Dispositions of Case Codes and Outcome Rates for Surveys*. Ann Arbor, MI: AAPOR.
- Atkinson, D., Schwanz, D., and Sieber, W.K. 1999. "Reporting Sources of Error in Analytic Publications." *Seminar on Interagency Coordination and Cooperation*. Washington, DC: U.S. Office of Management and Budget (Statistical Policy Working Paper 28). 329–341.
- Atrostic, B.K. and Burt, G. 1999. "Household Nonresponse: What We have Learned and a Framework for the Future." *Seminar on Interagency Coordination and Cooperation*. Washington, DC: U.S. Office of Management and Budget (Statistical Policy Working Paper 28). 153–180.
- Brick, J.M. and Kalton, G. 1996. "Handling Missing Data in Survey Research." *Statistical Methodology in Medical Research*. 5: 215–238.
- Council of American Survey Research Organizations. 1982. *On the Definition of Response Rates*. Port Jefferson, NY.
- Chakrabarty, R.P. and Torres, G. 1996. *American Housing Survey: A Quality Profile*. Washington, DC: U.S. Department of Housing and Urban Development and U.S. Department of Commerce.
- David, M., Little, R.J.A., Samuhel, M.E., and Triest, R.K. 1986. "Alternative Methods for CPS Income Imputation." *Journal of the American Statistical Association*. 81: 29–41.
- Dillman, D. 1978. *Mail and Telephone Surveys: The Total Design Method*. New York: John Wiley & Sons.
- Dillman, D. 1991. "The Design and Administration of Mail Surveys." *Annual Review of Sociology*. 17: 225–249.
- Federal Committee on Statistical Methodology. 1988. *Quality in Establishment Surveys*. Washington, DC: U.S. Office of Management and Budget (Statistical Policy Working Paper 15).
- Gonzalez, M., Kasprzyk, D., and Scheuren, F. 1995. "Exploring Nonresponse in U.S. Federal Surveys." *Seminar on New Directions in Statistical Methodology*. Washington, DC: U.S. Office of Management and Budget (Statistical Policy Working Paper 23). 603–624.
- Groves, R.M. 1989. *Survey Errors and Survey Costs*. New York: John Wiley & Sons.
- Groves, R.M. and Couper, M.P. 1998. *Nonresponse in Household Interview Surveys*. New York: John Wiley & Sons.
- Groves, R.M. and Couper, M.P. 1995. "Theoretical Motivation for Post-survey Nonresponse Adjustment in Household Surveys." *Journal of Official Statistics*. 11: 93–106.

Guenther, P.M. and Tippet, K.S. (eds.). 1993. *Evaluation of Nonresponse in the Nationwide Food Consumption Survey, 1987–88*. Washington, DC: U.S. Department of Agriculture, Human Nutrition Information Services. Nationwide Food Consumption Survey 1987–88 (NFCS Report No. 87-M-2).

Harris-Kojetin, B.A. and Robison, E. 1998. “Evaluating Nonresponse Adjustment in the Current Population Survey (CPS) using Longitudinal Data.” *Proceedings of Statistics Canada Symposium 98: Longitudinal Analysis for Complex Surveys*. Ottawa: Statistics Canada.

Herzog, T. and Rubin, D. 1983. “Using Multiple Imputations to Handle Nonresponse in Sample Surveys.” In Madow et al. (Eds.) *Incomplete Data in Sample Surveys*. 2: 209–245. New York: Academic Press.

Ingels, S.J. 1996. *Sample Exclusion in NELS:88, Characteristics of Base Year Ineligible Students: Changes in Eligibility Status after Four Years*. Washington, DC: U.S. Department of Education, National Center for Education Statistics (NCES 96–723).

Jabine, T. 1994. *Quality Profile for SASS: Aspects of the Quality of Data in the Schools and Staffing Surveys*. Washington, DC: U.S. Department of Education, National Center for Education Statistics (NCES 94–340).

Johnson, A.E., Botman, S.L., and Basiotis, P. 1994. “Nonresponse in Federal Demographic surveys: 1981–1991.” *Proceedings of the Section on Survey Research Methods*. Alexandria, VA: American Statistical Association. 983–988.

Kalton, G. and Kasprzyk, D. 1982. “Imputing for Missing Survey Responses.” *Proceedings of the Section on Survey Research Methods*. Alexandria, VA: American Statistical Association. 146–151.

Kalton, G. and Kasprzyk, D. 1986. “The Treatment of Missing Survey Data.” *Survey Methodology*. 12: 1–16.

Kasprzyk, D. and Kalton, G. 1997. “Measuring and Reporting the Quality of Survey Data.” *Proceedings of Statistics Canada Symposium 97: New Directions in Surveys and Censuses*. Ottawa: Statistics Canada. 179–184.

Kulka, R. 1995. “The Use of Incentives to Survey ‘Hard-to-Reach’ Respondents: A Brief Review of Empirical Research and Current Research Practices.” *Seminar on New Directions in Statistical Methodology*. Washington, DC: U.S. Office of Management and Budget (Statistical Policy Working Paper 23). 256–299.

Kydonieffs, L. and Stanley, J. 1999. “Establishment Nonresponse: Revisiting the Issues and Looking into the Future.” *Seminar on Interagency Coordination and Cooperation*. Washington, DC: U.S. Office of Management and Budget (Statistical Policy Working Paper 28). 181–227.

Lepkowski, J. 1989. “Treatment of Wave Nonresponse in Panel Surveys.” In D. Kasprzyk, G. Duncan, G. Kalton, and M.P. Singh (eds.), *Panel Surveys*. New York: John Wiley & Sons. 348–374.

- Lepkowski, J., Kalton, G., and Kasprzyk, D. 1989. "Weighting adjustments for partial nonresponse in the 1984 SIPP panel." *Proceedings of the Section on Survey Research Methods*. Alexandria, VA: American Statistical Association. 296–301.
- Lessler, J. and Kalsbeek, W. 1992. *Nonsampling Error in Surveys*. New York: John Wiley & Sons.
- Little, R.A. and Rubin, D. 1987. *Statistical Analysis with Missing Data*. New York: John Wiley & Sons.
- Loft, J., Riccobono, J., Whitmore, R., Fitzgerald, R., and Berkner, L. 1995. *Methodology Report for the National Postsecondary Student Aid Study*. Washington, DC: U.S. Department of Education, National Center for Education Statistics (NCES 95–211).
- Lyberg, L. and Dean, P. 1992. "Methods for Reducing Nonresponse Rates: A Review." Paper prepared for presentation at the 1992 meeting of the American Association for Public Opinion Research.
- Madow, W., Nisselson, H., and Olkin, I. 1983. *Incomplete Data in Sample Surveys*. New York: Academic Press.
- Montaquila, J. and Brick, J.M. 1997. *Unit and Item Response Rates, Weighting, and Imputation Procedures in the 1996 National Household Education Survey*. Washington, DC: U.S. Department of Education, National Center for Education Statistics (Working Paper No. 97–40).
- Moonesinghe, R., Mitchell, S., and Pasquini, D. 1995. "An Identification Study of Nonrespondents to the 1993 Survey of Doctorate Recipients." *Proceedings of the Section on Survey Research Methods*. Alexandria, VA: American Statistical Association. 453–458.
- Mosher, W., Judkins, D., and Goksel, H. 1989. "Response Rates and Non-response Adjustment in a National Survey." *Proceedings of the Section on Survey Research Methods*. Alexandria, VA: American Statistical Association. 273–278.
- National Science Foundation. 1999. *Graduate Students and Postdoctorates in Science and Engineering: Fall 1997 (Detailed Statistical Tables)*. Project Officer: J. Burrelli. Arlington VA (NSF 99–325).
- Oh and Scheuren, F. 1983. "Weighting Adjustment for Unit Nonresponse." In Madow et al. (eds.) *Incomplete Data in Sample Surveys*. 2: 143–184. New York: Academic Press.
- Osmint, J.B., McMahon, P.B., and Ware-Martin, A. 1994. "Response in Federally Sponsored Establishment Surveys." *Proceedings of the Section on Survey Research Methods*, Alexandria, VA: American Statistical Association. 977–982.
- Paxson, M.C., Dillman, D., and Tarnai, J. 1995. "Improving Response to Business Mail Surveys." In Cox et al. (eds.) *Business Survey Methods*. New York: John Wiley & Sons. 303–316.
- Rubin, D.B. 1987. *Multiple Imputation for Nonresponse in Surveys*. New York: John Wiley & Sons.

Rubin, D.B. 1996. "Multiple Imputation after 18+ Years." *Journal of the American Statistical Association*. 81: 366–374.

Scheuren, F., Monaco, D., Zhang, F., Ikosi, G., and Chang, M. 1996. *An Exploratory Analysis of Response Rates in the 1990–91 Schools and Staffing Survey (SASS)*. Washington, DC: U.S. Department of Education, National Center for Education Statistics (NCES 96–338).

Shettle, C.F., Guenther, P.M., Kasprzyk, D., and Gonzalez, M.E. 1994. "Investigating Nonresponse in Federal Surveys." *Proceedings of the Section on Survey Research Methods*. Alexandria, VA: American Statistical Association. 972–976.

Smith, T. Forthcoming. "Developing Nonresponse Standards." In R. Groves, (ed.), *Survey Nonresponse*. New York: John Wiley & Sons.

Spencer, B., Frankel, M., Ingels, S., Rasinski, K, and Tourangeau, R. 1990. *National Education Longitudinal Study of 1988: Base Year Sample Design Report*. Washington, DC: U.S. Department of Education, National Center for Education Statistics (NCES 90–463).

Tucker, C. and Harris-Kojetin, B.A. August 1998. "The Impact of Nonresponse on the Unemployment Rate in the Current Population Survey (CPS). In A. Koch and R. Porst (eds.), *Nonresponse in Survey Research: Proceedings of the Eighth International Workshop on Household Survey Nonresponse*. Mannheim, Germany: ZUMA. 45–54.

U.S. Energy Information Administration. 1996. *Residential Energy Consumption Survey Quality Profile*. Washington, DC: U.S. Department of Energy.